



Hierarchical Clustering Transcript

Hello everyone, welcome to lesson 1 video 2. In this video, we will study hierarchical clustering method. This method builds clusters step by step, either by merging smaller groups or by splitting larger ones. Now that we have known about clustering, let's learn about hierarchical clustering.

So basically clustering objects based on some hierarchy. So, one of the methods for doing hierarchical clustering is agglomerative clustering, wherein you bottom up the cluster, so let's say you start from divided data points and then you group together similar ones, one by one. So the first step is, start by treating each data points as its own cluster.

They are distinct clusters. Let's say if you have hundreds of data points, you have 100 clusters. Then what you do is, you find two closest based on some method of similarity, let's say distance or something, and then you merge them together into a new cluster.

And then you keep repeating the entire step number 2 until all the data points have been clubbed together. This process ends as all data points have been merged into one single giant cluster. So how does it look like? It actually looks like this.

You have, let's say for example you have 5 different data points, and you could because A and B are similar, you club them together, okay, you have one particular group. You have C, D, E, initially you have 5 data points and 5 groups. You could club A and B together, similarly you could club D and E together, and then once they are a group, you can club together D, E, and C, and you have A and B as one group on the left hand side.

Then you go up, eventually you group together, after you group together C and the group of D, E, you group them with A and B, right. So this way, the algorithm ends at once you have only one single group. Okay.

Now, this is measures of cluster distance. A key question which comes up is, how do we measure the distance between two clusters, right, which may contain multiple points or multiple data points. This is determined by thing which is called as linkage criteria.

You could have a single linkage, just like here, distance is equal to closest points. Let's say for example, you have closest points here, right. Whatever the distance between them, if that distance is as small, you could link them together.

So let's say this is a cluster A, this is cluster B, you have five data points here, you have five data points here, and you find distance between the closest data points. Hence, you club them together because you find that they are very, very nearby. Now you can also link together based on complete linkage.

So less distance between the farthest data points, that's what also you can use to link two groups. Similarly, you could also use averages, right, not distance between closest



and farthest, but also distance between the average. Well, let's say you take distance between the centroid, you average all these data points together, which are part of purple group, average all the data points which are a part of orange group, and then find the distance between them.

So that's what is average linkage. Now once you do all of that, you would get a situation like this. So if you start from the bottom, you have five groups, right, then you keep on forming group, now you have four groups, or you keep on combining groups, right.

Now you have three groups, then you combine them together, you have two groups, and finally when you at the top, you combine everything, you have one complete cluster. So this is a very simple method which is called as agglomerative clustering. This entire plot is called as dendrogram.

This helps us to visualise the data. I hope this is clear to everyone. This is one type of clustering.

We will also explore other types of clustering in the next video. In summary, hierarchical clustering provides a clear and interpretable structure for grouping data. The dendrogram allows us to visualise how clusters form at each and every stage.

In the next video, we will explore k-means clustering, one of the most powerful clustering methods that we use in industry today.

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